BTS–I–11.19–1448

Reg. No.

B

B.Tech. Degree I Semester Examination November 2019

CS/EC/IT 19-200-0102 B ENGINEERING PHYSICS

(2019 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A

(Answer ***ALL*** questions)

$(8 \times 3 = 24)$

I. (a) Explain the recording of the hologram.

(b) What is the basic principle of fibre optics?

(c) What are the important space lattices in the cubic system? Give their characteristics?

(d) Mention any three properties of liquid crystals.

(e) Write a note on fullerence.

(f) Explain the concept of penetration depth in superconductivity.

(g) State and explain Uncertainity Principle in quantum mechanics.

(h) The intensity of sound produced by thunder is 0.1 W/m^2. Calculate the intensity level in decibels.

PART B

$(4 \times 12 = 48)$

II. (a) Explain with a neat diagram, the construction and working of He-Ne laser. (8)

(b) What fraction of sodium atom is in the first excited state in a sodium vapour lamp at a temperature of 250^0 C? (4)

OR

III. (a) Define numerical aperture and derive an expression for numerical aperture and acceptance angle of fibre. (8)

(b) A step index fibre has n1= 1.68 , n2=1.44 . Calculate critical angle, maximum angle of refraction, acceptance angle and numerical aperture. (4)

IV. (a) What are Miller Indices? How will you determine the Miller indices of a given plane? (8)

(b) In a simple cubic crystal, find the ratio of intercepts on the three axes by (123) plane and also find the ratio of spacing of (110) and (111) planes. (4)

OR

V. (a) Write an essay about the properties of metallic glasses and their applications. (8)

(b) What is shape memory effect? (4)

VI. (a) Explain any four properties of nanomaterials in detail. (8)

(b) What is quantum confinement effect in nanomaterials? (4)

OR

VII. (a) Describe BCS theory briefly? (8)

(b) For a specimen, the critical fields are 1.4×10^{5} A/m and 4.2×10^{5} A/m for 14.2 K and 13.7 K. Calculate the critical temperature and critical magnetic field at 0 K and 4.2 K. (4)

VIII. (a) Derive Schrodinger's time dependent wave equation. (8)

(b) Calculate the velocity and kinetic energy of an electron of wavelength 1.66×10^{-10} m. (4)

OR

IX. (a) What is magnetostriction effect? Describe the production of ultrasonic waves by magnetostriction method. (8)

(b) A quartz crystal of thickness 2mm is vibrating at resonance. Calculate the fundamental frequency and first overtone. Given Young's modulus for quartz 7.9×10^{10} N/m^2 and density for quartz 2650 kg/m^3. (4)

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